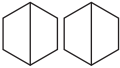


ANALAB CAPSULE

Tramadol hydrochloride



COMPOSITION:
Each capsule contains: Tramadol hydrochloride 50 mg

PRODUCT DESCRIPTION:
White granules containing in green-yellow opaque hard gelatin No.4 capsules printing in black BIOLAB's logo and "BIOLAB" on the surface.

ACTIONS:
Tramadol is a centrally acting opioid analgesic. It is a nonselective pure agonist at μ , δ and K opioid receptors with a higher affinity for the μ receptor. Other mechanisms which contribute to its analgesic effect are inhibition of neuronal re-uptake of norendrine and enhancement of serotonin release. Tramadol has an antitussive effect. In contrast to morphine, analgesic doses of tramadol over a wide range have no respiratory depressant effect. Also, gastrointestinal motility is not affected. Effects on the cardiovascular system tend to be slight. The potency of tramadol is reported to be 1/10-1/6 that of morphine. The onset of the analgesic effect of tramadol hydrochloride is rapid and the effect lasts for several hours. Tramadol hydrochloride is a potent drug and should not be used for minor pain.
It does not have an anti-inflammatory effect.

Tramadol hydrochloride has an analgesic effect in all the usual test models for centrally acting analgesics. On parenteral administration, the intensity of the effect is equivalent to that of other opioid analgesics, e.g. pentazocine and pethidine. On oral administration, tramadol hydrochloride is practically as effective as on parenteral application. The onset of activity is rapid and the duration of the effect is similar to that of morphine. Tramadol hydrochloride is a pure opiate agonist. The morphine antagonist, naloxone, neutralizes the analgesic effect of tramadol hydrochloride.

Pharmacokinetics:
More than 90% of tramadol is absorbed after oral administration. The mean absolute bioavailability is approximately 70%, irrespective of the concomitant intake of food. The difference between absorbed and non-metabolised available tramadol is probably due to the low first-pass effect. The first-pass effect after oral administration is a maximum of 30%.
Tramadol has a high tissue affinity (V_d 8- 203+40 l). It has a plasma protein binding of about 20%.
Following a single oral dose administration of tramadol 100 mg as capsules or tablets to young healthy volunteers, plasma concentrations were detectable within approximately 15 to 45 minutes within a mean C_{max} of 280 to 208 mcg/L and T_{max} of 1.6 to 2 h.
Tramadol passes the blood-brain and placental barriers. Very small amounts of the substance and derivative are found in the breast-milk.
Elimination half-life $t_{1/2\beta}$ is approximately 6 h, irrespective of the mode of administration. In patients above 75 years of age it may be prolonged by a factor of approximately 1.4.
Tramadol is mainly metabolised by means of N-and O-demethylation and conjugation of the O-demethylation products with glucuronic acid. Only O-desmethyltramadol is pharmacologically active. There are considerable interindividual quantitative differences between the other metabolites. So far, eleven metabolites have been found in the urine. The inhibition of one or both types of the isoenzymes CYP3A4 and CYP2D6 involved in the biotransformation of tramadol may affect the plasma concentration of tramadol or its active metabolite. Up to now, clinically relevant interactions have not been reported.
Tramadol and its metabolites are almost completely via the kidneys. Cumulative urinary excretion is 90% of the total radioactivity of the administered dose, in cases of impaired hepatic and renal function the half-life may be slightly prolonged.
In patients with cirrhosis of the liver, elimination half-lives of 13,3+4.9 h (tramadol) and 18,5+9.4 h (O-desmethyltramadol), in an extreme case 22,3 h and 36 h respectively, have been determined. In patients with renal insufficiency (creatinine clearance < 5 ml/min) the values were 11+3.2 h and 16,9+3 h, in an extreme case 19,5 h and 43,2 h respectively.
Tramadol has a linear pharmacokinetic profile within the therapeutic dosage range.
The relationship between serum concentrations and the analgesic effect is dose-dependent, but varies considerably in isolated cases. A serum concentration of 100-300 ng/ml is usually effective.

INDICATIONS:
Treatment of moderate to severe acute and chronic pain and in painful diagnostic measure and surgery.

DOSAGE AND ADMINISTRATIONS:
Oral, 50 mg
The dosage should be adjusted to the intensity of the pain, unless otherwise prescribed, ANALAB CAPSULE should be taken as follows independent of meals:

- **Single dose:**
For adults and adolescents (12 years and older) :
ANALAB CAPSULE is not approved for use in patients below 12 years old.
One 50 mg capsule to be taken with a little liquid. This is usually sufficient to relieve the pain. If, however, pain relief is unsatisfactory, a further 50 mg capsule may be taken after about 30-60 minutes.
- **In general the daily dose:**
Should not exceed 400 mg ANALAB CAPSULE.
In impaired renal or hepatic function it may be necessary to adjust the dose.
- **Paediatric population:**
The safety and efficacy of ANALAB CAPSULE has not been studied in the paediatric population. Therefore, use of ANALAB CAPSULE is not recommended in patients under 12 years of age.
- **Duration of treatment:**
During long-term treatment with ANALAB CAPSULE, the possibility of dependence cannot be entirely excluded. Therefore the physician is to decide on the duration of treatment and whether the preparation is to be withdrawn temporarily. ANALAB CAPSULE should not be given for longer than therapeutically necessary.

CONTRAINDICATIONS:
It is generally contraindicated in the following conditions:

- respiratory depression, especially in the presence of cyanosis and excessive bronchial secretion.
- in the presence of acute alcoholism, head injuries and conditions in which intracranial pressure is raised.
- it should not be used in patients receiving MAO inhibitors.
- previous hypersensitivity to ANALAB CAPSULE
- it should not be used during pregnancy.
- it should not be administered during lactation as tramadol and its metabolites have been detected in breast milk.
- Children younger than 18 years to treat pain after surgery to remove the tonsils and/or adenoids.
- Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

PRECAUTIONS AND WARNINGS:

- It should be used with extreme caution in patients with the following conditions:
Decreased respiratory reserves; and should not be given during an attack of bronchial asthma or in heart failure secondary to chronic lung disease, hypothyroidism, adrenocortical insufficiency, impaired kidney or liver function, prostatic hypertrophy, shock or inflammatory or obstructive bowel disorders, myasthenia gravis.
- It should be given with great care to infants, especially neonates.
- The administration of opioid analgesics during labour may cause respiratory depression in the newborn infant.
- Even when administered according to instructions, the preparations may affect the reaction ability of the patients to such extent that his capacity to drive or operate machines may be impaired. This applies particularly in conjunction with alcohol.
- Paediatric population:
The safety and efficacy of ANALAB CAPSULE has not been studied in the paediatric population. Therefore, use of ANALAB CAPSULE is not recommended in patients under 12 years of age.
- Respiratory depression :
Administer ANALAB CAPSULE cautiously in patients at risk for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia, hypercapnia, or pre-existing respiratory depression, as in these patients, even therapeutic doses of ANALAB CAPSULE may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioid analgesics should be considered. When large doses of tramadol are administered with an anesthetic medication or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.
- Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism : Some individuals may be CYP2D6 ultra-rapid metabolizers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1). This rapid conversion could result in higher than expected opioid-like side effects including life-threatening respiratory depression. The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African, Americans, and 16-28% in North Africans, Ethiopians, and Arabs. Data are not available for other ethnic groups.
- Information for the patients:
ANALAB CAPSULE is a potent drug for the relief of pain, e.g. in wound pain, fractures, severe nerve pain, tumor pain, heart attack. It should not be used for minor pain. The effect sets in quickly and lasts for some hours.
- Risks from Concomitant Use with Benzodiazepines :
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of ANALAB CAPSULE with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, prescribing of concomitant dosing of these drugs for use in patients for whom alternative treatment options are inadequate. If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.
If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.
If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.
Follow patients closely for signs and symptoms of respiration depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when ANALAB CAPSULE is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined.
Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (see DRUG INTERACTIONS).
- Serotonin Syndrome with Concomitant Use of Serotonergic Drugs:
Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of ANALAB CAPSULE with serotonergic drugs (See DRUG INTERACTIONS). This may occur within the recommended dosage range.
Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See DRUG INTERACTIONS). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue ANALAB CAPSULE if serotonin syndrome is suspected.
- Adrenal Insufficiency:
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure.
If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement doses of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.
- Sexual Function/Reproduction:
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)
- Use in pregnancy :** Tramadol has been shown to cross the placenta. There are no adequate and well-controlled studies in pregnant women. Safe use in pregnancy has not been established. ANALAB CAPSULE is not recommended for pregnant women.
- Use in lactation :** Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400 mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason tramadol should not be used during lactation or alternatively, breast-feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

SIDE EFFECTS:
There have been occasional reports of nausea, sweating, dry mouth, dizziness, lightheadedness, headache, motorial weakness, changes in appetite, blurred vision, micturition disorders, psychic effect, allergic reactions, worsening asthma, epileptiform convulsion, increase intracranial pressure and bradycardia, dependence, increase liver enzymes, affect reactions.
In very rare cases, various psychic side effects may occur depending on personality and duration of treatment. These include changes in mood, changes in activity and changes in cognitive and sensorial capacity (e.g. decision behavior, perception disorder).
Allergic reactions (e.g. dyspnoea, bronchospasm, wheezing, angioneurotic oedema) and anaphylaxis have also been reported.
In rare cases, there may be effects on cardiovascular regulation (palpitation, tachycardia, postural hypotension up to cardiovascular collapse). These side effects may occur particularly on IV administration and in patients under physical strain.
Moreover, headache, retching, vomiting, constipation, gastrointestinal irritation (e.g. feeling of pressure in the stomach, bloating) and dermal reactions (e.g. pruritus, exanthema) may occur.
Respiratory depression (rare)
Postmarketing Experience:
Serotonin syndrome (See PRECAUTIONS AND WARNINGS)
Adrenal insufficiency (See PRECAUTIONS AND WARNINGS)
Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.
Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

DRUG INTERACTIONS:

- It should be given with extreme caution in patients taking monoamine oxidase inhibitors or within 14 days of stopping such treatment.
- The depressant effects of ANALAB CAPSULE are enhanced by depressants of the central nervous system such as alcohol, anaesthetics, hypnotics and sedatives, tricyclic antidepressants and phenothiazines.
- Opioid analgesics with some antagonist activity such as buprenorphine, butorphanol, nalbuphine or pentazocine may precipitate withdrawal symptoms in patients who have recently used pure agonists such as ANALAB CAPSULE.
- On the concomitant administration of ANALAB CAPSULE with substances which also act on the central nervous system (e.g. tranquilizers, hypnotics) the sedative effects (fatigue) may be intensified. At the same time, however, combining ANALAB CAPSULE with a tranquilizer, for example, will probably have a favourable effect on the pain sensation.
ANALAB CAPSULE should not be used in patients receiving MAO inhibitors.
- Benzodiazepines :
Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.
The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.
Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see PRECAUTIONS AND WARNINGS). Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.
- Serotonergic Drugs:
The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue ANALAB CAPSULE if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See PRECAUTIONS AND WARNINGS).

OVERDOSAGE:
Clinical features: Cold, clammy skin; confusion; convulsion; severe drowsiness; severe drowsiness; pinpoint pupils of eyes; slow heartbeat; slow or troubled breathing; unconsciousness; severe weakness.
Treatment: Intensive supportive therapy may be required to correct failure and shock. In addition, the specific antagonist naloxone hydrochloride is used to counteract very rapidly the severe respiratory depression and coma produced by excessive doses of opioid analgesics.

STORAGE:
Store in a dry place at temperature not exceeding 30°C.
Keep medicine out of reach of children.
The shelf-life period is 3 years.

SUPPLY:
Capsule: Blister pack 10x10's

Date of revision : December 2017

Note : #0-ไม่เกี่ยวข้องเป็น "ANALAB CAPSULE" #1-ไม่เกี่ยวข้อง #2-เกี่ยวข้องบางส่วนหรือ "PRECAUTION, DRUG INTERACTION" #3-ไม่เกี่ยวข้อง #4-ไม่เกี่ยวข้อง #5-เกี่ยวข้องบางส่วนหรือ "SIDE EFFECT" #6-เกี่ยวข้องบางส่วนหรือ "Pregnancy, side effect, drug interaction" #7-เกี่ยวข้อง "Yes" #8-เกี่ยวข้องบางส่วนหรือ "Pregnancy, side effect, drug interaction" #9-ไม่เกี่ยวข้อง		RA BIOLAB	
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